

## CO 250 Review 9: CS conditions and applications

### Complementary slackness conditions (CS conditions)

Another way to characterize optimal solutions using duality.

For a primal-dual pair with feasible solutions  $x, y$ , the CS conditions are:

- $x_i = 0$  or the  $i$ -th dual constraint is tight (holds with equality)
- $y_j = 0$  or the  $j$ -th primal constraint is tight.

Example:  $\max (3 \ -4 \ 1)x$   
s.t.  $\begin{pmatrix} 3 & -1 & 2 \\ 1 & 2 & 3 \end{pmatrix} x \leq \begin{pmatrix} 5 \\ -3 \end{pmatrix}$   
 $x_1 \geq 0, x_2 \leq 0, x_3 \text{ free}$

$$\min (5 \ -3)y$$

s.t.  $\begin{pmatrix} 3 & 1 \\ -1 & 2 \\ 2 & 3 \end{pmatrix} y \leq \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}$   
 $y_1 \geq 0, y_2 \leq 0$

CS conditions:

- $x_1 = 0$  or  $3y_1 + y_2 = 3$
- $x_2 = 0$  or  $-y_1 + 2y_2 = -4$
- $y_1 = 0$  or  $3x_1 - x_2 + 2x_3 = 5$
- $y_2 = 0$  or  $x_1 + 2x_2 + 3x_3 = -3$

CS theorem: For  $\bar{x}, \bar{y}$  feasible for (P), (D),  $\bar{x}, \bar{y}$  are optimal for (P), (D) if and only if  $\bar{x}, \bar{y}$  satisfy all CS conditions.

Example: Verify that  $\bar{x} = (0, -3, 1)^T$ ,  $\bar{y} = (2, -1)^T$  are optimal using CS conditions.

First check they are feasible. ✓

Check all 4 conditions:

$$\bullet \bar{x}_1 = 0$$

$$\bullet -\bar{y}_1 + 2\bar{y}_2 = -4$$

$$\bullet 3\bar{x}_1 - \bar{x}_2 + 2\bar{x}_3 = 5$$

$$\bullet \bar{x}_1 + 2\bar{x}_2 + 3\bar{x}_3 = -3$$

✓

⇒ optimal

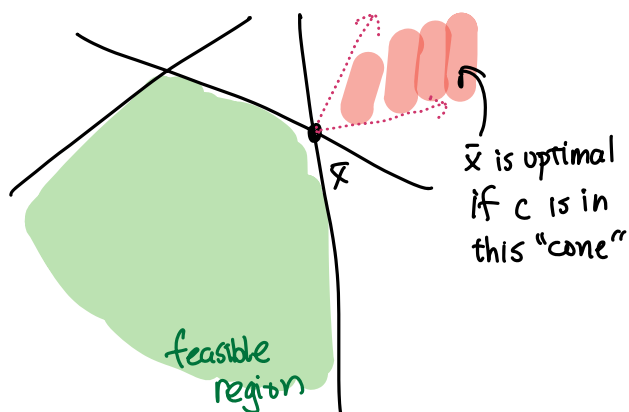
Note: ① Sometimes it is easier to apply equivalent conditions: "if a variable is not zero, then the corresponding constraint must be tight" or "if a constraint is not tight, then the corresponding variable must be zero".

② It is possible that both conditions hold (these are inclusive "or"s).

Exercise: Use CS conditions and feasibility to show that there are no other optimal solutions in the example above.

## Application 1: Geometry of optimality

Given an LP  $\max \{c^T x : Ax \leq b\}$  and a feasible solution  $\bar{x}$ , which  $c$  makes  $\bar{x}$  optimal?



The cone generated by  $v^1, \dots, v^k$  is  
 $\{\lambda_1 v^1 + \dots + \lambda_k v^k : \lambda_1, \dots, \lambda_k \geq 0\}$ .

Theorem:  $\bar{x}$  is optimal if and only if  $c$  is in the cone generated by the normal vector of the tight constraints for  $\bar{x}$ .

Proof sketch:

$$\begin{array}{ll} \max & (c^T) x \\ \text{s.t.} & \begin{pmatrix} A \end{pmatrix} x \leq \begin{pmatrix} b \end{pmatrix} \end{array} \rightarrow \begin{array}{ll} \min & (b^T) y \\ \text{s.t.} & \begin{pmatrix} A^T \end{pmatrix} y = (c) \\ & y \geq 0. \end{array}$$

$\bar{x}$  optimal  $\Rightarrow$  CS conditions hold.

Rewrite dual constraints:

Non-tight constraint for  $\bar{x}$   
 $\Rightarrow y_j = 0$ .

$$\underbrace{\begin{pmatrix} \quad \end{pmatrix} y_1 + \cancel{\begin{pmatrix} \quad \end{pmatrix} y_2} + \dots + \cancel{\begin{pmatrix} \quad \end{pmatrix} y_m}}_{\text{tight constraints}} = \begin{pmatrix} c \end{pmatrix}$$

$\Rightarrow c$  is in the cone of tight constraints.

## Application 2: Farkas' Lemma (existence of a certificate of infeasibility)

Farkas' Lemma: Exactly one of the following conditions hold:

• There exists  $x$  s.t.  $Ax = b, x \geq 0$

$\leftarrow$  the system  $Ax = b, x \geq 0$  is feasible

• There exists  $y$  s.t.  $A^T y \geq 0, b^T y < 0$

$\leftarrow$  there is a certificate of infeasibility

Proof sketch: ① Show that both conditions cannot hold simultaneously. (algebra)

② Assume that  $x$  does not exist. Then  $\max \{ 0^T x : Ax = b, x \geq 0 \}$  is infeasible.

The dual is  $\min \{ b^T y : A^T y \geq 0 \}$ . This is either infeasible or unbounded.

But  $y = 0$  is feasible, so it must be unbounded. So there exists  $y$  such that  $b^T y < 0$  and  $A^T y \geq 0$ .